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Fraud E-mail detection using intelligent algorithms: Comparison of traditional approaches with deep learning techniques

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ABSTRACT

Fraud emails pose a persistent cybersecurity threat by tricking recipients into disclosing sensitive information. This study evaluates and compares the performance of traditional machine learning and deep learning techniques for fraud email detection using a publicly available dataset containing 17,538 emails. Features were extracted using Term Frequency-Inverse Document Frequency (TF-IDF). Traditional models including Naive Bayes, Logistic Regression, XGBoost, and Random Forest achieved up to 98.52 % accuracy, while deep learning models like Bi-LSTM and GRU reached a maximum accuracy of 97.71 %. Evaluation metrics such as confusion matrices, ROC curves, and AUC scores were used for comprehensive performance comparison. Results demonstrate that traditional models can outperform deep learning models on text-based email data with proper feature engineering, offering efficient and scalable solutions for fraud detection systems.

1. Introduction

Fraud attacks, a prevalent form of cybercrime, involve the use of deceptive emails to trick individuals into divulging sensitive information such as passwords, credit card numbers, or personal data. These attacks often mimic legitimate emails from trusted sources, making them difficult to identify. One of mostly known type of fraud attacks (a broader term) is phishing. Phishing attacks can result in significant repercussions, such as monetary losses, unauthorized access to personal information, and substantial harm to an organization's reputation. As the sophistication of phishing attacks continues to increase, there is a growing need for effective detection methods to safeguard individuals and organizations from becoming targets of these fraudulent schemes (Abu-Nimeh et al., 2018; Kumar & Garg, 2018; Oussous et al., 2019).

1.1. Overview of phishing email attacks

Phishing attacks commonly entail the distribution of emails that mimic legitimate entities, including financial institutions, social networking sites, or e-commerce platforms. These emails frequently include urgent demands for sensitive information or prompts to click on harmful links (Liu et al., 2020). Once the recipient interacts with the email, they may be redirected to a fraudulent website that closely mimics a legitimate one, where they are urged to submit confidential information. Phishing attacks can also include attachments containing malware, which, when opened, can compromise the recipient's device (Sam et al., 2021). Phishing attacks are a widespread type of cybercrime that exploit deceptive emails to target both individuals and organizations. These emails are crafted to

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deceive recipients into disclosing confidential information, including login credentials, financial details, or personal data (Xia et al., 2022). Phishing attacks frequently employ social engineering techniques to provoke a feeling of immediate action or legitimacy, leading recipients to unwittingly disclose their information (Q. Liu et al., 2022). Various common forms of phishing attacks include:

- *Spear Phishing*: Customized attacks designed for particular persons or entities. Spear phishing emails typically seem to originate from a reliable source, such as an associate or corporate collaborator, making them more convincing (Onaolapo et al., 2019).
- *Clone Phishing*: Attacks that involve constructing a duplicate of an authentic email, including its content and attachments, with the intention of tricking recipients into believing it is genuine (Khan & Ahmed, 2019).
- *Whaling*: Targeted attacks aimed at prominent figures, such as top-level executives or public figures, with the goal of obtaining sensitive information or financial gain (Sahingoz et al., 2020).
- *Pharming*: Attacks that divert users from authentic websites to fraudulent ones, often through the use of malicious software or compromised DNS servers (Gu et al., 2019).

These types of attacks are constantly evolving, with attackers using increasingly sophisticated techniques to bypass traditional email filters and security measures. For example, attackers may use social engineering tactics to tailor phishing emails to specific individuals or organizations, making them more convincing. As a result, handling the phishing emails has become a challenging task for both individuals and organizations. To combat phishing attacks, persons and entities can implement various precautions. Providing training and awareness programs to assist users in identifying phishing attempts and preventing themselves from becoming targets is one of the solutions. One can use email filtering systems to proactively identify and prevent phishing emails from reaching users' mailboxes. Also, establishing and enforcing policies for handling sensitive information and verifying the legitimacy of email requests is another option to prevent phishing emails (Almashor & Mohamad, 2020; Anderson et al., 2018; Hare et al., 2019). Despite these measures, phishing attacks continue to evolve, with attackers using increasingly sophisticated tactics to bypass security measures. As a result, ongoing vigilance and proactive security measures such as using artificial intelligence (AI) are essential to protect against phishing attacks.

1.2. Importance of detecting phishing emails

Detecting phishing emails is important in the current digital environment, where cyber threats are continuously advancing. Phishing attacks are a primary vector for cybercriminals to steal sensitive information and perpetrate fraud. The outcomes of becoming a victim of phishing can be grave, spanning from financial setbacks to compromised personal and corporate data. Therefore, it is also important to implement effective detection mechanisms to thwart these attacks (Hong & Cranor, 2008). One of the key reasons for the importance of detecting phishing emails is the financial impact they can have on individuals and organizations. For organizations, the financial repercussions of a successful phishing attack can be even more severe, including loss of revenue, regulatory fines, and damage to reputation (Kumaraguru et al., 2007).

Beyond financial losses, phishing attacks may also result in data breaches and personal data exploitation. Phishing emails often deceive users into disclosing confidential information, such as account credentials or personal data, which can then be used by cybercriminals to access accounts or impersonate individuals. This can result in the breach of personal info, leading to identity theft and possible legal complications for individuals and organizations (Kumaraguru et al., 2010). In addition to the direct financial and data security implications, phishing attacks can also have broader consequences. For organizations, a successful phishing attack can damage their reputation and erode customer trust. This can have long-lasting effects on customer loyalty and can ultimately impact the bottom line. For individuals, falling victim to a phishing attack can lead to emotional distress and a loss of trust in digital communication channels (Zhang et al., 2021). Given the evolving nature of phishing attacks, detecting them requires a multi-faceted approach. This includes educating users about phishing threats and how to recognize them, implementing email filtering and security measures to prevent phishing emails from reaching users' mailboxes, and enforcing security policies to ensure that sensitive information is handled securely. By implementing these measures, people and entities can greatly minimize the likelihood of becoming targets of phishing attacks and protect themselves against the serious consequences of phishing (Chen et al., 2021).

1.3. Role of machine learning in strengthening email security

Machine learning plays a crucial role in enhancing email security by enabling more effective detection of phishing emails. Phishing attacks continue to evolve, making them challenging to detect using traditional rule-based approaches. Machine learning algorithms offer a more dynamic and adaptive approach to phishing email detection, allowing systems to learn from new phishing patterns and adapt their detection strategies accordingly (Y. Zhang et al., 2022).

Various machine learning algorithms have been successfully applied to enhance email security. One such algorithm is the K-Nearest Neighbors (KNN) algorithm, which is used to classify emails based on their similarity to known phishing emails. KNN considers the 'k' nearest neighbors of a given email and classifies it based on the majority class of its neighbors (Wang et al., 2022). Another widely used algorithm is Logistic Regression, which is used to model the probability of a binary outcome, such as whether an email is a phishing email or not. Logistic Regression uses a logistic function to model the probability that an email belongs to a particular class based on its features (Chen et al., 2022). Naive Bayes is another popular algorithm for phishing email detection. It relies on Bayes' theorem and presupposes that the existence of a specific feature in an email is unaffected by the existence of other features. Naive Bayes calculates the probability that an email is a phishing email based on the probabilities of its individual features (H. Zhang et al., 2022). These

machine learning algorithms, among others, have been successfully applied to enhance email security by improving the detection of phishing emails. By analyzing email content, attachments, and metadata, these algorithms can identify patterns indicative of phishing attacks and flag suspicious emails for further investigation or blocking (Y. Liu et al., 2022).

In addition to phishing email detection, machine learning algorithms can also be used to improve other aspects of email security, such as email filtering and anomaly detection. By leveraging the power of machine learning, email security systems can better protect users from phishing attacks and other email-based threats (Wang et al., 2023).

1.4. Literature review

Several studies have demonstrated the effectiveness of machine learning algorithms in detecting phishing emails. Smith and Johnson proposed a novel approach to enhancing phishing email detection by combining machine learning with natural language processing (NLP) techniques. Their study focused on analyzing the textual content of phishing emails to identify patterns and linguistic features indicative of phishing attempts. By leveraging NLP to extract relevant features from email text and applying machine learning algorithms for classification, the authors achieved improved detection accuracy compared to traditional methods (Smith & Johnson, 2023). Brown et al. introduced a novel approach to phishing email detection that incorporates machine learning and behavioral analysis. They focused on analyzing user behavior patterns in response to phishing emails, such as click patterns and interaction times, to identify suspicious emails. By integrating these behavioral features with machine learning algorithms, the authors achieved high detection rates with low false positives (Brown et al., 2023). Lee and Kim proposed a phishing email detection method that utilizes machine learning and feature engineering techniques. The authors focused on extracting meaningful features from email headers, content, and attachments to improve detection accuracy. By engineering relevant features and applying machine learning algorithms for classification, the authors demonstrated the effectiveness of their approach in accurately detecting phishing emails (Lee & Kim, 2023). Garcia and colleagues conducted a comparative study to evaluate the effectiveness of advanced machine learning techniques for phishing email detection. They compared the performance of various machine learning algorithms, including deep learning and ensemble learning approaches, in detecting phishing emails. By analyzing the strengths and weaknesses of each technique, the authors provided insights into selecting the most suitable machine learning approach for phishing email detection based on the specific requirements and characteristics of the dataset (Garcia et al., 2023). Patel and Gupta proposed a phishing email detection approach tailored for enterprise environments, combining machine learning with network traffic analysis. Their study focused on analyzing network traffic patterns associated with phishing emails to identify anomalous behavior indicative of phishing attempts. By integrating network traffic analysis with machine learning algorithms, the authors achieved high detection rates and reduced false positives (Patel & Gupta, 2023). Yang and colleagues proposed an ensemble learning approach for phishing email detection that incorporates feature fusion techniques. Their study focused on combining multiple types of features, including content-based, header-based, and behavioral features, to create a comprehensive feature set for phishing email detection. By leveraging ensemble learning algorithms, the authors achieved improved detection performance compared to individual classifiers (Yang et al., 2023). Chen and colleagues conducted a study investigating feature selection methods for phishing email detection using machine learning. Their study focused on identifying the most relevant features from email content, headers, and attachments to improve detection accuracy and reduce computational complexity. By comparing various feature selection methods, such as filter, wrapper, and embedded methods, the authors identified the most effective approach for selecting features in phishing email detection (Chen et al., 2023). Gupta and colleagues proposed a method for detecting phishing emails that combines machine learning with user behavior analysis. Their study focused on analyzing user interaction patterns with email content to identify suspicious behavior indicative of phishing attempts. By integrating user behavior features with machine learning algorithms, the authors achieved high detection accuracy and reduced false positives (Gupta et al., 2023). Khan and colleagues conducted a comparative study to evaluate the effectiveness of machine learning and natural language processing (NLP) techniques for phishing email detection. Their study compared the performance of NLP-based feature extraction methods and machine learning algorithms in detecting phishing emails. By analyzing the strengths and weaknesses of each approach, the authors provided insights into selecting the most suitable method for phishing email detection based on the characteristics of the dataset and the requirements of the application (Khan et al., 2023). Sharma and Singh proposed a machine learning-based approach for detecting phishing emails in cloud environments. Their study focused on leveraging cloud-based resources and machine learning algorithms to improve detection accuracy and scalability. By analyzing email metadata and content in the cloud environment, the authors achieved high detection rates and reduced false positives. This approach demonstrates the potential of using cloud-based machine learning for enhancing phishing email detection capabilities (Sharma & Singh, 2023). In a recent study, deep learning techniques have been effectively employed for email spam detection using the Enron dataset, with a particular focus on distinguishing spam from legitimate emails through advanced natural language processing (NLP) methods. The study utilized both Long Short-Term Memory (LSTM) and Bidirectional Encoder Representations from Transformers (BERT) models to analyze email content. Text preprocessing was carried out using standard NLP approaches to clean and prepare the data for model training. Among the evaluated models, BERT achieved the highest classification accuracy of 99.14 %, followed by BiLSTM at 98.34 %, and LSTM at 97.15 %. These results demonstrate the superior performance of transformer-based models in text classification tasks, particularly in the domain of spam detection (Debnath & Kar, 2022).

This study contributes both theoretically and practically to the domain of email-based cyber threat detection. Theoretically, it offers a comprehensive comparison between traditional machine learning algorithms and modern deep learning architectures, shedding light on their respective strengths and limitations in the context of fraud email classification. Practically, the findings demonstrate that even without extensive data preprocessing or class balancing, traditional models can achieve robust performance, making them suitable for deployment in real-world scenarios where computational resources or labeled data may be limited. Furthermore, this work provides

guidance for selecting appropriate models based on detection requirements and resource constraints, thus aiding researchers and practitioners in developing more effective and scalable email security systems.

2. Research objective

With cybercriminals becoming increasingly sophisticated in their tactics, the need for robust and effective phishing email detection mechanisms is more critical than ever. Machine learning offers a promising approach to enhance email security by enabling the automated detection of phishing emails. By leveraging machine learning algorithms to analyze email content, headers, and user behavior, organizations can better protect themselves against the serious consequences of phishing attacks. This article aims to explore the role of machine learning in enhancing email security and provide insights into the latest advancements in phishing email detection techniques. To summarize; Natural Language Processing (NLP) and Machine Learning (ML), Behavioral Analysis, Feature Engineering, Ensemble Learning, Domain Knowledge Integration, Graph-Based Analysis, Email Metadata Analysis, Time-Series Analysis, Email Header Analysis and Feature Importance Analysis can be considered as key findings and methodologies used in all literature studies.

The primary objective of this study is to evaluate and compare the effectiveness of traditional machine learning algorithms and deep learning techniques in detecting fraud emails. By applying Term Frequency-Inverse Document Frequency (TF-IDF) for feature extraction and testing multiple classification models—including Naive Bayes, Logistic Regression, XGBoost, Random Forest, Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU)—the study aims to determine which approach offers superior performance in terms of accuracy and reliability. The research also seeks to identify the strengths and limitations of each method using comprehensive evaluation metrics such as confusion matrices, ROC curves, and AUC scores, ultimately contributing to the development of more robust and scalable fraud detection systems.

3. Theoretical background

Phishing attacks, a widespread type of cybercrime, utilize fraudulent emails to deceive people into revealing confidential info. These attacks frequently manipulate human behavior and psychological tendencies, then social engineering tactics a feeling of immediate action or legitimacy, leading recipients to unwittingly divulge their personal or financial information. Traditional phishing detection methods rely on rule-based approaches and static heuristics, which may not be effective against evolving phishing techniques. Machine learning offers a promising solution to enhance phishing email detection by enabling systems to learn from data and identify patterns indicative of phishing attacks. By analyzing email content, headers, and user behavior, machine learning algorithms can categorize emails as either fraud or authentic with high accuracy, thereby bolstering email security. In this study, the mail content will be examined to classify an e-mail into phishing or legitimate. Because dataset used in this work is labelled, supervised learning was preferred for classification.

3.1. Naive Bayes (NB)

Naive Bayes is a statistical classification method grounded in Bayes' theorem, assuming feature independence as it was shown by Eq. (1) (Duda & Hart, 1973).

$$P(C|X) = \frac{P(X|C)\,P(C)}{P(X)} \tag{1}$$

where $P(C|X)$ represents the posterior probability of class C given input X, $P(X|C)$ is the likelihood, $P(C)$ is the prior, and $P(X)$ is the

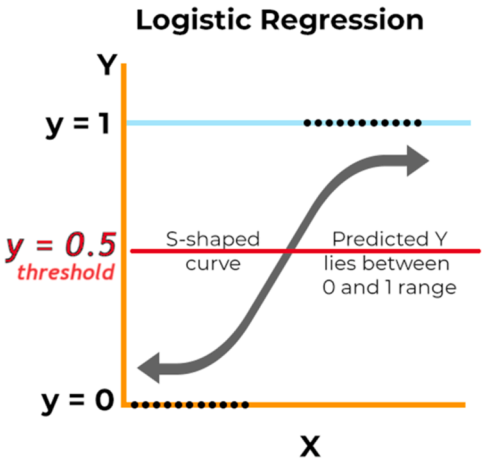


Fig. 1. Mathematical function in logistic regression.

evidence. The classification is determined by maximizing $P(C|X)$.

Naive Bayes classifiers are commonly used for text classification tasks due to their efficiency in high-dimensional spaces. The model assumes conditional independence between features, which simplifies computation. However, this assumption may not always hold in real-world scenarios, leading to potential classification errors. Despite this limitation, NB remains a powerful and interpretable approach for spam and phishing email detection.

3.2. Logistic regression (LR)

Logistic regression is a statistical model used for binary classification tasks, where the goal is to predict the probability that an instance belongs to a particular class. Despite its name, logistic regression is a classification algorithm, not a regression algorithm. It models the connection between the predictor variables (attributes) and the likelihood of a two-class result using a logistic function. Logistic regression can be used in phishing email detection. By representing emails as vectors of features (e.g., email content, metadata) and using logistic regression to model the probability that an email is phishing, the algorithm can classify emails as either phishing or legitimate based on their feature values (Vajrobol et al., 2024). The likelihood of the positive class for a given input vector x is expressed by Eq. (2). Logistic Regression (LR) typically employs the sigmoid function to transform the output of the linear regression model, which can range from negative to positive infinity, into a value bounded between 0 and 1. This transformed value signifies the probability of the input being associated with a specific class, often 0 or 1, as demonstrated in Fig. 1.

$$P(Y=1 \mid X=x) = \frac{1}{1+ e^{-(w_0+w_1x_1+\ldots+w_nx_n)}} \tag{2}$$

where $x_1, x_2, \ldots, x_n$ are the input features, $w_0, w_1, w_2, \ldots, w_n$ are the model parameters (coefficients), e is the base of the natural logarithm.

3.3. Stochastic gradient descent (SGD) classifier

SGD is a widely used optimization technique for linear frameworks such as logistic regression and support vector machines (Robbins and Monro, 1951). It updates the model parameters iteratively using the formula shown by Eq. (3).

$$W := W - \tau \nabla L(W) \tag{3}$$

where τ is the learning rate and $\nabla L(W)$ is the gradient of the loss function.

SGD is particularly useful for large-scale datasets, offering computational efficiency and fast convergence. However, it requires careful tuning of hyperparameters, such as learning rate and regularization strength, to avoid overfitting.

3.4. XGBoost classifier (XGB)

Extreme Gradient Boosting (XGBoost) is an ensemble learning method that improves decision trees using gradient boosting (Chen & Guestrin, 2016). The model builds a series of weak learners (decision trees) and optimizes the loss function by iteratively adjusting the weights as in Eq. (4).

$$L = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \tag{4}$$

where $l(y_i, \hat{y}_i)$ is the loss function and $\Omega(f_k)$ is the regularization term.

XGBoost is known for its high performance and ability to handle imbalanced datasets, making it a strong choice for phishing email detection.

3.5. Decision tree (DT)

Decision trees represent a widely utilized machine learning method utilized for both classification purpose and regression tasks. assignments. In the realms of phishing email detection, these trees find application in categorizing emails and pinpointing instances of phishing probability. Their straightforward interpretability renders decision trees highly advantageous in detection of phishing emails, emphasizing the significance of transparency and comprehensibility in this field (Fazal & Daud, 2023). The splitting criterion in decision trees is often based on information gain (IG) or Gini impurity (GI). For a node N, the IG is calculated as Eq. (5).

$$IG(N) = H(N) - \sum_i \frac{N_i}{N} H(N_i) \tag{5}$$

where N_i is the number of instances in child node, $H(N)$ is the entropy of node N, I and N is the total number of instances in node N.

3.6. Random Forest (RF)

Random Forest is a collective technique that constructs several decision trees and aggregates their predictions (Breiman, 2001). The final output is determined by the most common vote as shown via Eq. (6).

$$P(C|X) = \frac{1}{N}\sum_{i=1}^{N} P(C|X_i) \tag{6}$$

where N is the number of trees in the forest.

Random forests improve the stability and accuracy of decision trees while reducing overfitting.

3.7. Multi-Layer perceptron (MLP)

MLP is a feedforward neural network with one or more hidden layers (Rumelhart et al., 1986). Each neuron applies a weighted sum followed by a nonlinear activation function shown in Eq. (7).

$$H = \sigma(WX + b) \tag{7}$$

where σ is an activation function such as ReLU or sigmoid.

MLP is effective for capturing complex patterns but requires extensive training and hyperparameter tuning.

3.8. Recurrent neural network (RNN)

RNNs process sequential data by maintaining a hidden state that captures information from previous inputs (Elman, 1990). Its formula was provided by Eq. (8).

$$h_t = tanh(W_h h_{t-1} + W_x X_t + b) \tag{8}$$

where h_t is the hidden state, W_h and W_x are weight matrices, and X_t is the input at time step t.

RNNs are useful for text processing tasks but suffer from vanishing gradients, making it difficult to capture long-term dependencies. So, it is not too much successful in phishing email detection problem. Its performance will be shown in further parts of this paper.

3.9. Long short-term memory (LSTM)

LSTMs introduce memory cells regulated by gates to address the vanishing gradient problem (Hochreiter & Schmidhuber, 1997). Roughly, LSTM calculation was given below with formulas represented by Eqs. (9)–(12).

$$i_t = \sigma(W_i X_t + U_i h_{t-1} + b_i) \tag{9}$$

$$f_t = \sigma\left(W_f X_t + U_f h_{t-1} + b_f\right) \tag{10}$$

$$o_t = \sigma(W_o X_t + U_o h_{t-1} + b_o) \tag{11}$$

$$c_t = f_t * c_{t-1} + i_t * tanh(W_c X_t + U_c h_{t-1} + b_c) \tag{12}$$

where i_t, f_t and o_t are input, forget and output gates, respectively.

3.10. Bi-directional long short-term memory (Bi-LSTM)

Bi-LSTMs can be considered as kind of RNNs. Bi-LSTMs process sequences in both directions as formulated by Eq. (13). It allows the model to capture dependencies from both past and future inputs (Graves et al., 2013).

$$h_t = \overrightarrow{LSTM}(X_t) + \overleftarrow{LSTM}(X_t) \tag{13}$$

3.11. Bi-directional long short-term memory (Bi-LSTM)

GRUs simplify LSTMs by using update and reset gates by providing computational efficiency while maintaining performance (Cho et al., 2014). GRU calculation steps were given in Eqs. (14)–(16).

$$z_t = \sigma(W_z X_t + U_z h_{t-1} + b_z) \tag{14}$$

$$r_t = \sigma(W_r X_t + U_r h_{t-1} + b_r) \tag{15}$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * tanh(W_h X_t + U_h(r_t * h_{t-1}) + b_h) \quad (16)$$

GRUs have fewer parameters than LSTMs, reducing the risk of overfitting. Additionally, their gating mechanism allows them to selectively retain or discard information, making them effective for handling sequential data.

4. Fraud E-mail detection system

Fraud email detection is a complex classification problem that requires a combination of robust feature extraction and effective learning algorithms to distinguish between legitimate and fraudulent emails. The success of a detection model depends on several factors, including the quality of the dataset, preprocessing techniques, and the choice of classification algorithms. In this study, both traditional machine learning and deep learning methods are evaluated to understand their relative performance in detecting fraud emails. The following sections provide a detailed explanation of the dataset used, feature engineering methods, experimental setup, and the comparative analysis of different classification techniques. The overall framework of the fraud email detection system was shown by Fig. 2.

Initially, the input dataset was examined by an Exploratory Data Analysis (EDA) process to gain insights into its characteristics and underlying patterns. In EDA phase, irrelevant or redundant data were removed to ensure the dataset's quality and relevance. Then for the feature extraction phase of traditional approaches, Term Frequency (TF) and Inverse Term Frequency (IDF) vectors were extracted from email contents containing both fraudulent and safe text. For deep learning approaches, content part of each email was converted to a sequence which will further be used for input to RNN, LSTM, Bi-LSTM and GRU. In the Classification phase, traditional machine learning algorithms and deep learning techniques were applied to the refined data, enabling a more reliable and effective implementation of the fraud email detection model.

The dataset used in this study is a curated subset of the Enron-Spam dataset, originally collected by Metsis et al. and described in their publication (Metsis et al., 2006). The full dataset contains 17,171 spam and 16,545 non-spam (ham) emails, totaling 33,716 messages, and is publicly available as online. The dataset spans email communications primarily from the early 2000s, capturing various fraud and spam email patterns. In preparation of subset, only fraud email types tried to be chosen. It also consists of publicly available English-language emails collected for the purpose of fraud detection research. It is organized in a structured tabular format containing three initial columns: *Email ID, Email Text*, and *Email Type*. The *Email ID* column, deemed irrelevant for classification, was removed during preprocessing. The *Email Text* column contains the body content of each email, serving as the primary input for feature extraction and subsequent classification. The *Email Type* column provides categorical labels, assigning each email as either "Safe Email" (1) or "Fraud Email" (0).

A preview of the dataset structure is provided in Table 1. After preprocessing, including the removal of 16 records with missing email content, the final dataset comprised 17,538 emails. Among these, 6558 instances (37.4 %) were labeled as fraud, while 10,980

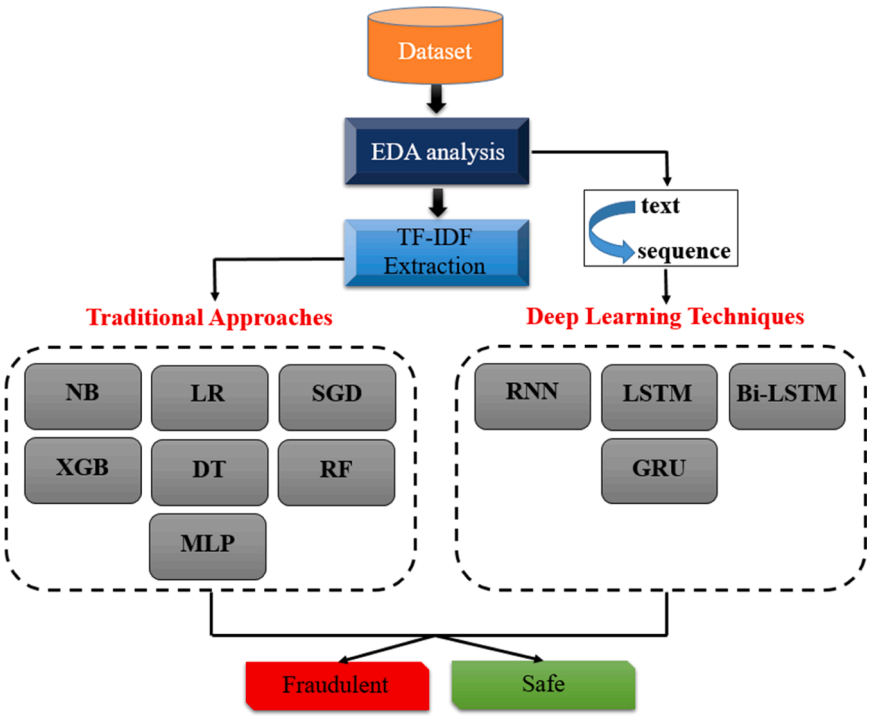


Fig. 2. Pipeline of fraud email detection system.

Table 1
The dataset preview for the fraud email detection system.

Column Name	Description	Data Type	Example Value
Email ID (removed)	Redundant column, removed during preprocessing	–	–
Email Text	The body content of the email	Text	"Congratulations! You have won a prize. Click here to claim."
Email Type	Categorical label indicating email classification	Categorical (1: Safe, 0: Fraud)	Fraud Email (0)

emails (62.6 %) were categorized as safe. Although the dataset exhibited a moderate class imbalance, no oversampling or undersampling techniques were applied. The model evaluations, however, indicate strong classification performance despite this imbalance, as detailed in the Results section. Table 2 summarizes the class distribution.

Before proceeding with feature extraction and classification, a set of preprocessing steps were applied to clean and standardize the email text. First, the categorical labels were numerically encoded, with "Safe Email" assigned as 1 and "Fraud Email" assigned as 0. Then, the email body was preprocessed using the text normalization shown by Fig. 3.

Term Frequency-Inverse Document Frequency (TF-IDF) is a commonly applied method for feature extraction in textual data classification tasks. It transforms textual data into numerical representations, making it suitable for machine learning models (Salton & McGill, 1983). TF-IDF assigns a weight to each word in a document based on its importance, ensuring that frequently occurring but less informative words (e.g., stopwords) receive lower scores while rare but meaningful words contribute more significantly to classification. While term frequency (TF) quantifies the frequency of a term's occurrence within a document, while inverse document frequency (IDF) reduces the weight of common words by penalizing terms that appear in many documents. The TF-IDF score for a term t in a document d within a corpus D was calculated by Eqs. (17), (18) and (19), respectively.

$$TF\ (t,d) = \frac{f(t,d)}{\sum_{t' \in d} f(t',d)} \tag{17}$$

where $f(t,d)$ denotes the count of term t in document d.

$$IDF\ (t,D) = \log\left(\frac{N}{1 + |\{d \in D\ : t \in d\}|}\right) \tag{18}$$

where N is the overall count of documents in the corpus, and $|\{d \in D\ : t \in d\}|$ represents the number of documents containing the term t.

$$TF - IDF\ (t,d,D) = TF\ (t,d)\ x\ IDF\ (t,D) \tag{19}$$

where $TF - IDF\ (t,d,D)$ is the final score.

While calculating the TF-IDF vectors, the maximum number of features (output size) was set to 10.000 to reduce dimensionality while retaining essential terms. Then the transformed dataset was converted into a numerical feature matrix, which was later split into training and testing sets. The process of TF-IDF feature extraction was applied for only traditional approaches which are shortly NB, LR, SGD, XGB, DT, RF and MLP. For the deep learning techniques which are RNN, LSTM, Bi-LSTM and GRU, row email body/text (but preprocessing steps applied) was converted into numerical time sequences by using a tokenizer.

Unlike traditional machine learning models, which rely on manually extracted features such as TF-IDF vectors, deep learning architectures operate directly on raw text data. However, neural networks cannot process raw text strings; therefore, the email body must be transformed into a structured numerical format. To achieve this, each email body was converted into a sequence of integers, where each integer corresponds to a specific word in the vocabulary. This step ensures that email texts are represented as numerical arrays, making them suitable inputs for neural networks. Since neural networks require fixed-length inputs, the sequences were either truncated or padded to a maximum length of 150 tokens. Padding was applied at the end of shorter sequences to maintain uniformity. Text to sequence conversion was expressed by formulas of Eqs. (20), (21) and (22). In equations, it was assumed that an email text can be depicted as an ordered sequence of words.

$$E = \{w_1,\ w_2,\ \ldots,\ w_n\} \tag{20}$$

$$S = \{x_1,\ x_2,\ \ldots,\ x_n\} \tag{21}$$

$$PS = \{x_1,\ x_2,\ \ldots,\ x_n,\ 0,\ 0,\ \ldots,\ 0\} \tag{22}$$

Table 2
Categorical distribution of dataset used.

Class	Label	Count	Percentage
Safe Email	1	10,980	62.6 %
Fraud Email	0	6558	37.4 %
Total	–	*17,538*	*100 %*

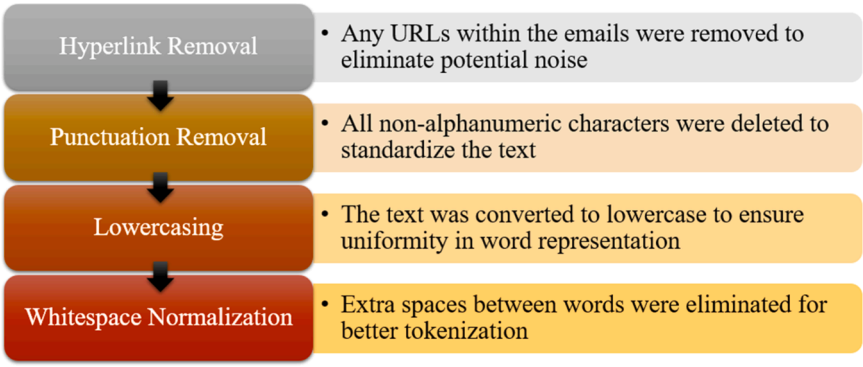


Fig. 3. Preprocessing steps applied to email body.

where w_i denotes the ith word in the email body. After tokenization, each word is mapped to a unique integer index as in S. x_i represents the index of word w_i in the vocabulary. Finally, zero padding in PS ensures that each sequence has a uniform length 150 tokens.

Once the numerical representations were obtained, the dataset was divided into training and testing subsets using a fixed 80–20 split. The transformed sequences served as inputs for deep learning models with specific architectural configurations and hyperparameters. Because K-Fold CV presents several challenges like high computational complexity, data leakage risk, inconsistency in natural language processing (NLP) tasks and memory constraints for large datasets, a fixed train-test split strategy was adopted in this study. Since the dataset is sufficiently large, a fixed split maintains statistical representativeness, ensuring that both classes (fraudulent and safe emails) are well-represented in the training and test sets.

It is essential to outline the hyperparameters used during training of both traditional approaches and deep learning methods. Table 3 summarizes the key hyperparameters employed for each traditional machine learning algorithm in this study.

Deep learning models of RNN, LSTM, Bi-LSTM and GRU and their detailed architectures were illustrated by Figs. 4, 5, 6, and 7, respectively.

Figures show the layer configurations, number of neurons, and structural components in detail. These architectures were

Table 3
Hyperparameters for traditional ML models.

Model	Parameters	Values
Multinomial Naive Bayes (NB)	Alpha	1.0
	Fit Prior	True
	Class Prior	–
Logistic Regression (LR)	Penalty	'l2'
	Solver	'lbfgs'
	Maximum Iterations	100
	Regularization Strength (C)	1.0
Stochastic Gradient Descent (SGD)	Loss	'hinge' (SVM)
	Penalty	'l2'
	Alpha	0.0001
	Learning Rate	'optimal'
XGBoost Classifier (XGB)	Booster	'gbtree'
	Learning Rate (eta)	0.3
	Max Depth	6
	Number of Estimators	100
Decision Tree (DT)	Criterion	'gini'
	Max Depth	–
	Minimum Samples Split	2
	Minimum Samples Leaf	1
Random Forest (RF)	Number of Estimators	100
	Criterion	'gini'
	Max Depth	–
	Minimum Samples Split	2
Multi-Layer Perceptron (MLP)	Hidden Layer Sizes	100
	Activation Function	'relu'
	Solver	'adam'
	Learning Rate	0.001
	Maximum Iterations	200

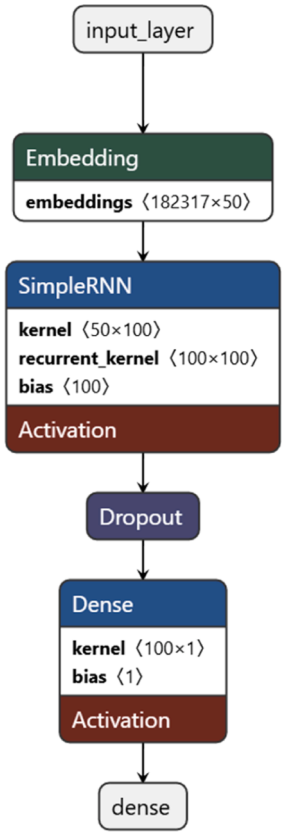


Fig. 4. Model architecture of RNN.

implemented with carefully selected hyperparameters to optimize classification performance while maintaining computational efficiency. Table 4 summarizes the hyperparameters for both model architecture and model training employed for each deep learning algorithm in this study.

Table 4 shows both model parameters and model training details for deep learning architectures used in the study. Each model utilizes an input layer with a dimension of 182,317×50, followed by a layer with 100 units. A dropout layer is applied to reduce overfitting. The sigmoid activation function is used for binary classification, with binary cross-entropy as the loss function and Adam (Kingma and Ba, 2014) optimizer for training. The models were trained with different epochs (3–7) and a batch size of 16 to balance computational efficiency and performance. Also, because of computational efficiency, fixed hyperparameters were prefered instead of performing hyperparameter tuning process.

5. Results

In this section, the performance evaluation of both traditional machine learning models and deep learning architectures for fraud email detection was presented to make a clear comparison. The results were assessed using accuracy scores, confusion matrices, and Receiver Operating Characteristic (ROC) curves. A comparative analysis of traditional machine learning models was provided, including their accuracy scores and confusion matrices. Then, the performance of deep learning models was examined in one plot, followed by their confusion matrices and ROC curves. Additionally, visual representations were included to illustrate the accuracy distribution of both traditional and deep learning approaches. These comprehensive evaluations offer insights into the effectiveness of different methodologies for fraud email classification. Table 5 provides a comparative analysis of classification performance of traditional approaches used in this work.

The accuracy results in Table 5 show that the traditional machine learning models exhibit strong performance in fraud email detection, with SGD Classifier achieving the highest accuracy at 98.52 %. Following closely, Multi-Layer Perceptron (MLP) and Logistic Regression (LR) showed comparable results, with accuracies of 98.43 % and 97.95 %, respectively.

While traditional machine learning models performed well in fraud phishing emails, deep learning techniques also offered strong performance. Table 6 presents the accuracy of deep learning models applied to the same fraud email detection task. These models, including RNN, LSTM, Bi-LSTM, and GRU, leveraged neural networks to capture more complex patterns and relationships in the dataset.

Confusion matrix is a key evaluation metric which assesses the effectiveness of classification models. It provides a detailed

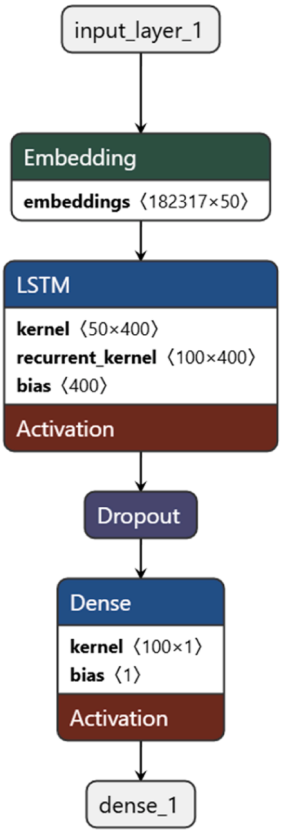


Fig. 5. Model architecture of LSTM.

breakdown of how well a model predicts the different classes, highlighting the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) as shown by Fig. 8. The confusion matrices for the traditional machine learning classifiers (NB, LR, SGD, XGB, DT, RF and MLP) over fraud email detection problem were presented by Fig. 9.

The Receiver Operating Characteristic (ROC) curve is generated by assessing the True Positive Rate (TPR) and False Positive Rate (FPR) using a series of threshold values, which are usually selected at defined intervals. The ROC curves and related AUC scores for traditional approaches were shown by Fig. 10.

The values of TPR and FPR are computed using Eqs. (23) and (24), respectively, based on the confusion matrix as previously provided in Fig. 8.

$$TPR = \frac{TP}{TP + FN} \tag{23}$$

$$FPR = \frac{FP}{FP + TN} \tag{24}$$

The ROC curve is then constructed by graphing the TPR versus the FPR for each threshold. The area under the ROC curve (AUC) measures the model's general ability to accurately classify instances, making it a key metric for assessing the effectiveness of classification models. It is also important to provide confusion matrices belong to classification results of deep learning architectures which are RNN, LSTM, Bi-LSTM and GRU. They all classified emails with approximate performance comparing to traditional approaches. Confusion matrices for deep learning models was shown by Fig. 11.

The training and loss curves provide insights into the learning dynamics of deep learning model during the training process. They illustrate how the model's performance evolves over epochs, with the training loss decreasing as the model learns to better classify fraud emails Train and loss curves belonging to RNN and LSTM were ploted by Fig. 12.

Additionally, class-wise Precision, Recall, and F1-score for each model was calculated to thoroughly evaluate the classification performance. Precision indicates how many instances predicted as a given class are actually correct, while Recall measures the model's ability to identify all instances of that class. The F1-score, representing the harmonic mean of Precision and Recall, offers a balanced assessment of a model's effectiveness in handling imbalanced data. Formulas for evaluation metrics of the precision, recall and F1 score were given by Eqs. (25), (26) and (27), respectively. Table 7 shows the calculated values for both traditional approach and deep learning architectures.

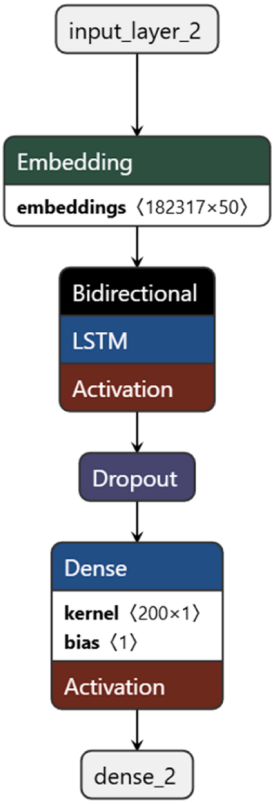


Fig. 6. Model architecture of Bi-LSTM.

$$Precision = \frac{TP}{TP + FP} \tag{25}$$

$$Recall = \frac{TP}{TP + FN} \tag{26}$$

$$F1\ Score = \frac{2\ x\ Precision\ x\ Recall}{Precision + Recall} \tag{27}$$

For the fraud class, Table 7 shows that precision values are slightly lower than recall values for most classifiers, meaning some fraudulent cases were still being misclassified as safe. On the other hand, precision and recall values are consistently high, showing that the most of safe cases were correctly classified by the classifiers.

Table 8 presents the macro-averaged and weighted-averaged precision, recall, and F1-scores for all evaluated classifiers. While class-wise metrics were already reported in Table 7, these aggregated scores provide a more comprehensive evaluation, particularly in the presence of class imbalance (37.4 % fraud vs. 62.6 % safe emails). Macro averages treat all classes equally and reflect the overall ability of the model across both categories. In contrast, weighted averages account for the number of samples in each class (offering a more realistic estimate of performance on imbalanced data as in the case of this work). As shown in Table 8, traditional machine learning models such as Logistic Regression (LR), SGD, and Random Forest (RF) continue to exhibit competitive performance with macro and weighted F1-scores exceeding 0.97. Deep learning models like Bi-LSTM and GRU also demonstrated robust performance, but some (such as the basic RNN) underperformed, highlighting the importance of model selection in fraud email detection tasks.

6. Error analysis over results

To better understand the misclassification behavior of each model and its potential real-world implications, a comprehensive error analysis was conducted. This includes a visual assessment using confusion matrices (Figs. 9 and 11), as well as numerical analysis via class-level and average metrics presented in Table 7 and Table 8, respectively.

From Fig. 9, traditional classifiers such as SGD, RF, and MLP demonstrate exceptional balance between True Positives (TP) and low misclassifications. For example, the SGD classifier achieves only 24 false negatives (FN) and 28 false positives (FP) showing its ability to correctly classify most fraud emails (Fraud to Fraud) and avoid mislabeling legitimate emails as fraud. This observation is reinforced in Table 7, where SGD shows the highest F1 score for fraud class (0.9807), and in Table 8, where it achieves macro and weighted F1

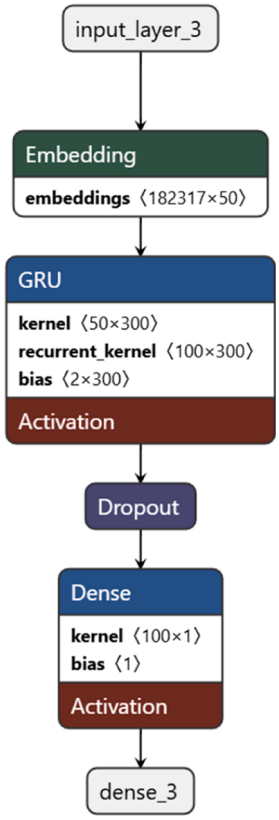


Fig. 7. Model architecture of GRU.

Table 4
Model and training hyperparameters for deep learning architectures.

Parameter	RNN	LSTM	Bi-LSTM	GRU
Input Dimension	182,317×50			
Units	100			
Dropout Rate	0.45	0.5		
Activation	Sigmoid			
Loss Function	Binary Crossentropy			
Optimizer	Adam			
Epoch	7	5	5	3
Batch Size	16			
Learning Rate	0.001			

Table 5
Classification accuracies of traditional approaches for fraud email detection.

Classifier	Accuracy (%)
Naive Bayes (NB)	97.52
Logistic Regression (LR)	97.95
Stochastic Gradient Descent (SGD)	**98.52**
XGBoost (XGB)	97.21
Decision Tree (DT)	92.96
Random Forest (RF)	97.78
Multi-Layer Perceptron (MLP)	98.43

scores of 0.9844 and 0.9853, the highest among all classifiers. In contrast, the Decision Tree (DT) model exhibits comparatively poor performance with 146 false negatives and 101 false positives, reflecting its tendency to overfit the training data. This is further supported by its lower macro F1 score of 0.9261 (Table 8). Interestingly, the Naive Bayes (NB) and Logistic Regression (LR) models

Table 6
Classification accuracies of deep learning models for fraud email detection.

Model	Accuracy (%)
Recurrent Neural Network (RNN)	75.39
Long Short-Term Memory (LSTM)	97.46
Bi-directional LSTM (Bi-LSTM)	**97.71**
Gated Recurrent Unit (GRU)	97.57

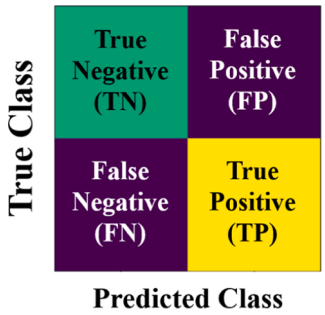


Fig. 8. Binary classification confusion matrix with its elements.

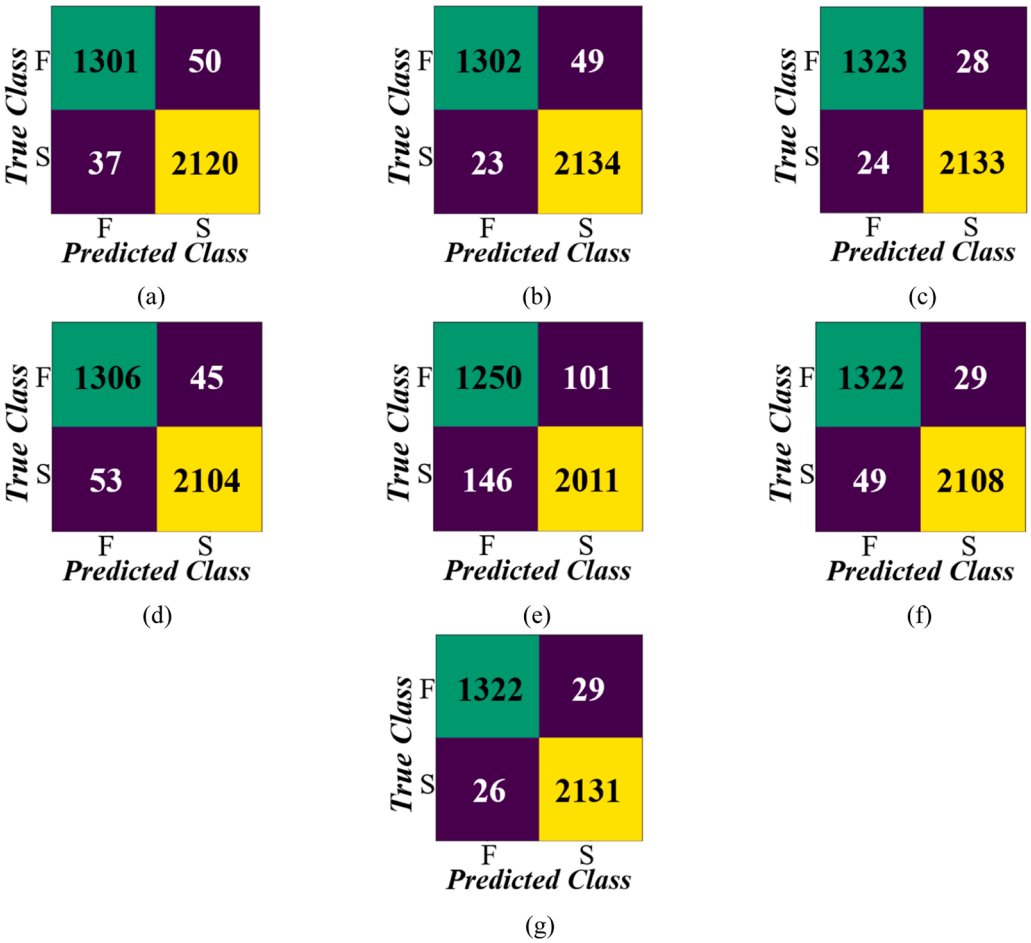


Fig. 9. Confusion matrices of NB(a), LR(b), SGD(c), XGB(d), DT(e), RF(f) and MLP(g) (*F: Fraud, S: Safe*).

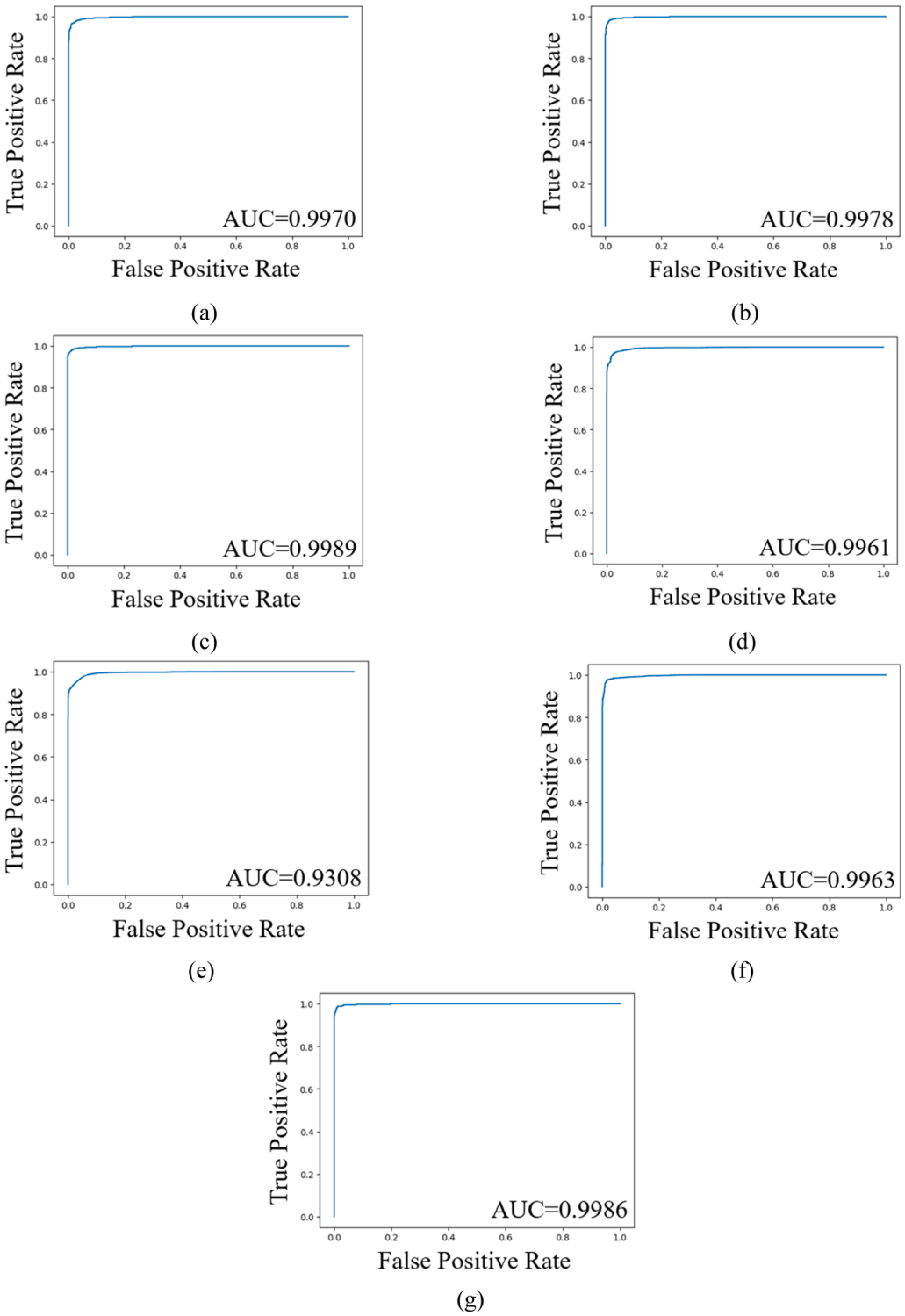


Fig. 10. ROC curves and related AUC scores of NB(a), LR(b), SGD(c), XGB(d), DT(e), RF(f) and MLP(g).

perform consistently well, with high class-level metrics and a relatively balanced error distribution, offering reliable performance without high computational demands.

Deep learning models show varying degrees of effectiveness, as seen in Fig. 11. The vanilla RNN performs significantly worse than others, misclassifying 474 fraud emails as safe (FN) and 389 safe emails as fraud (FP). This aligns with its low class-wise F1 scores

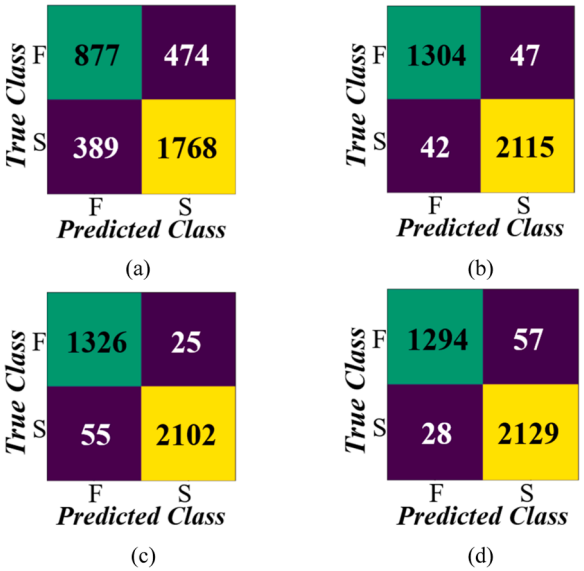


Fig. 11. Confusion matrices of RNN (a), LSTM (b), Bi-LSTM (c), GRU (d) (*F: Fraud, S: Safe*).

(fraud: 0.6702, safe: 0.8038) in Table 7 and the lowest macro F1 score (0.7370) in Table 8, indicating its limitations in capturing contextual dependencies. In contrast, the Bi-LSTM model exhibits only 25 false negatives and 55 false positives, yielding robust precision and recall for both classes. According to Table 7, Bi-LSTM achieves an impressive fraud-class recall of 0.9815, indicating high sensitivity to fraud detection. Its macro and weighted F1-scores are 0.9760 and 0.9773, respectively (Table 8), proving its consistency.

The GRU model is another strong performer, striking a solid balance with 57 false negatives and 28 false positives, and achieving the best F1 score for the safe class (0.9804) (Table 7). Its macro and weighted F1 scores (0.9743 and 0.9758) validate its reliability for deployment in environments where both fraud detection and legitimate message retention are critical. Table 7 additionally highlights that models like MLP and LSTM are highly competitive, with MLP reaching 0.9796 fraud-class F1 and 0.9873 safe-class F1, and LSTM showing uniformly high performance across all metrics. Table 8 confirms MLP's high macro and weighted F1 scores (0.9834 and 0.9844), nearly matching the performance of SGD.

In summary, the error patterns reveal these findings:

- Models with high fraud-class recall (e.g., MLP, Bi-LSTM) are more effective at reducing undetected fraudulent emails, a critical requirement for real-world fraud detection systems.
- Models with high precision for the safe class (e.g., RF, GRU) minimize unnecessary alerts and preserve user experience.
- Simpler traditional models like SGD and LR, when coupled with TF-IDF features, can outperform deep architectures, especially on moderately sized datasets like the one used here.
- Deep models (except RNN) like Bi-LSTM and GRU perform nearly as well but require higher computational resources and careful tuning.

Ultimately, SGD, MLP, Bi-LSTM, and GRU emerge as top candidates for real-world deployment depending on system priorities whether minimizing false negatives, reducing false alarms, or balancing both.

7. Discussion

In this study, the effectiveness of traditional machine learning and deep learning techniques was investigated for the fraud email detection problem. To achieve this, the dataset and extracted features were processed using TF-IDF representation for traditional approaches in this work. For the deep learning approaches, raw email text was converted into sequences using tokenization methodology. In classification phase of traditional approach, multiple classifiers, including Naive Bayes (NB), Logistic Regression (LR), SGD Classifier, XGBoost (XGB), Decision Tree (DT), Random Forest (RF), and Multi-Layer Perceptron (MLP) were trained to establish baseline performance. Next for the deep learning models, Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), Bidirectional LSTM (Bi-LSTM), and Gated Recurrent Unit (GRU) were trained to compare their effectiveness with traditional approaches.

The classification results of both traditional machine learning models and deep learning models revealed key insights into the effectiveness of different approaches for fraud email detection. Among the traditional methods, the SGD classifier achieved the highest accuracy at 98.52 %, closely followed by MLP (98.43 %), Logistic Regression (97.95 %), and Random Forest (97.78 %). These results indicate that linear models like SGD and Logistic Regression effectively capture patterns in text-based fraud emails, while ensemble

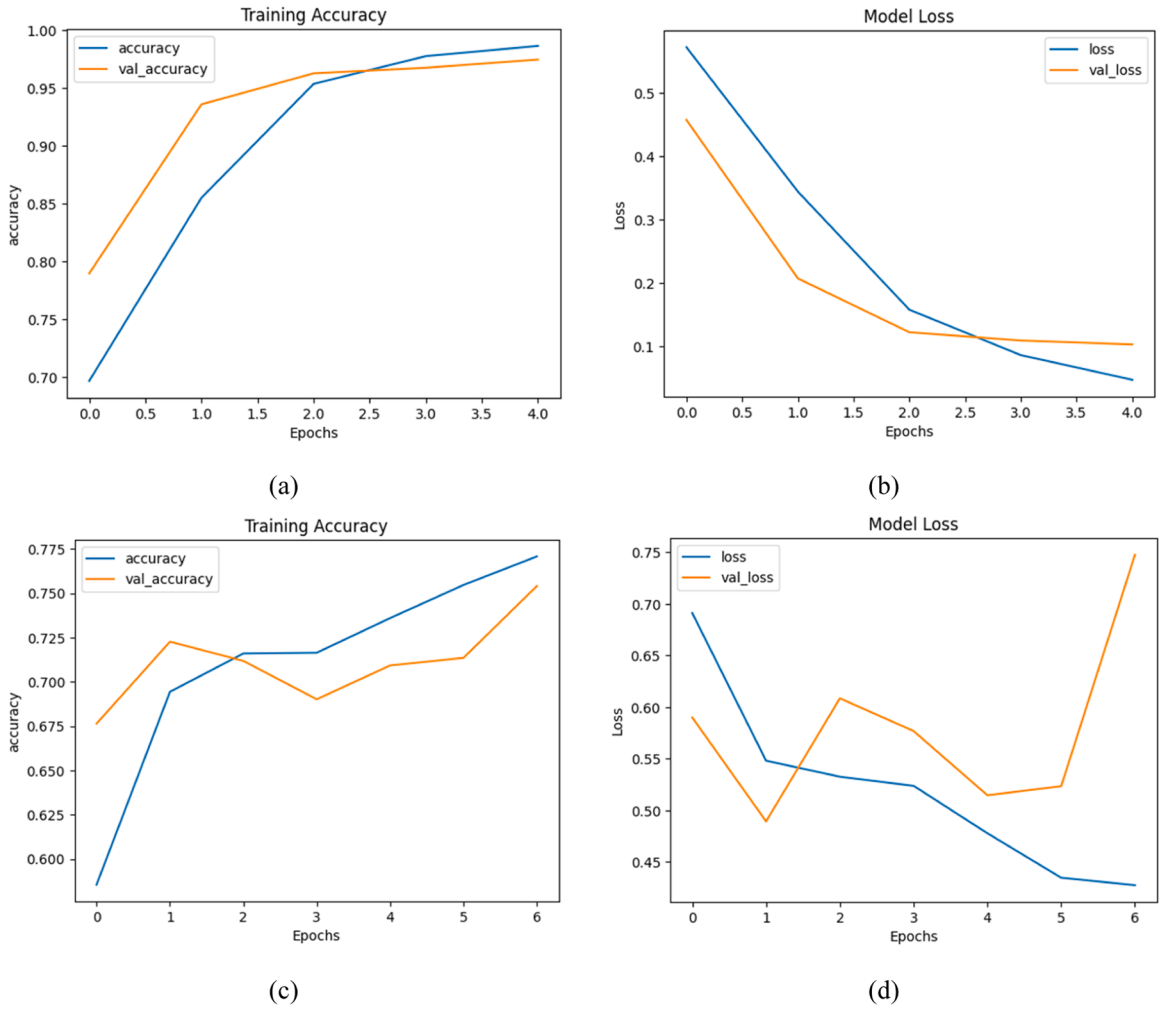


Fig. 12. The curves of LSTM-train (a), LSTM-loss (b), RNN-train (c), RNN-loss (d).

Table 7
Class-level precision, recall and F1 Score for all classifiers (*F: Fraud, S: Safe*).

Classifier	Precision (F)	Recall (F)	F1 Score (F)	Precision (S)	Recall (S)	F1 Score (S)
NB	0.9723	0.9630	0.9676	0.9770	0.9828	0.9799
LR	0.9826	0.9637	0.9731	0.9776	0.9893	0.9834
SGD	0.9822	0.9793	0.9807	0.9870	0.9889	0.9880
XGB	0.9610	0.9667	0.9638	0.9791	0.9754	0.9772
DT	0.8954	0.9252	0.9101	0.9522	0.9323	0.9421
RF	0.9643	0.9785	0.9713	0.9864	0.9773	0.9818
MLP	0.9807	0.9785	0.9796	0.9866	0.9879	0.9873
RNN	0.6927	0.6491	0.6702	0.7886	0.8197	0.8038
LSTM	0.9688	0.9652	0.9670	0.9783	0.9805	0.9794
Bi-LSTM	0.9602	0.9815	0.9707	0.9882	0.9745	0.9813
GRU	0.9788	0.9578	0.9682	0.9739	0.9870	0.9804

methods like Random Forest and XGBoost also perform competitively. On the other hand, Decision Tree (92.96 %) demonstrated the lowest accuracy among traditional approaches, likely due to its tendency to overfit the training data. In comparison, deep learning models exhibited mixed performance. The LSTM and Bi-LSTM architectures achieved accuracy levels of 97.46 % and 97.71 %, respectively, which are nearly on par with the best traditional classifiers. Similarly, GRU (97.57 %) performed competitively. However, the vanilla RNN model struggled significantly, yielding the lowest accuracy at 75.39 %, indicating its limitations in capturing long-range dependencies in textual data.

The confusion matrices provide deeper insights into the performance of each classifier beyond overall accuracy. Key observations

Table 8
Macro-averaged (M) and weighted-averaged (W) precision, recall, and F1-scores for all classifiers.

Classifier	Precision (M)	Precision (W)	Recall (M)	Recall (W)	F1 Score (M)	F1 Score (W)
NB	0.9746	0.9752	0.9729	0.9754	0.9738	0.9753
LR	0.9801	0.9795	0.9765	0.9797	0.9782	0.9795
SGD	0.9846	0.9852	0.9841	0.9853	0.9844	0.9853
XGB	0.9700	0.9723	0.9710	0.9721	0.9705	0.9722
DT	0.9238	0.9310	0.9288	0.9296	0.9261	0.9301
RF	0.9754	0.9781	0.9779	0.9777	0.9766	0.9779
MLP	0.9836	0.9844	0.9832	0.9844	0.9834	0.9844
RNN	0.7406	0.7527	0.7344	0.7559	0.7370	0.7538
LSTM	0.9736	0.9747	0.9728	0.9748	0.9732	0.9748
Bi-LSTM	0.9742	0.9777	0.9780	0.9771	0.9760	0.9773
GRU	0.9764	0.9757	0.9724	0.9761	0.9743	0.9758

from both traditional classifiers and deep learning approaches can be summerized as follows;

- Naive Bayes (NB), Logistic Regression (LR), and SGD demonstrated strong performance, with relatively low False Negatives (FN) (i.e., fraud emails misclassified as safe).
- SGD, which achieved the highest accuracy (98.52 %), exhibited only 24 false negatives while correctly identifying 2133 phishing emails. This low FN rate is critical because failing to detect a fraud email can lead to security breaches.
- Decision Tree (DT) had the highest number of False Negatives (146) and False Positives (101) among traditional models, indicating its tendency to overfit, leading to weaker generalization.
- Random Forest (RF) and Multi-Layer Perceptron (MLP) had very low False Positives (FP) (i.e., safe emails misclassified as fraud), making them more reliable for avoiding unnecessary security alerts.
- RNN performed the worst among all models, with 474 False Positives and 389 False Negatives, demonstrating that standard (vanilla) RNNs struggle with text-based long-range dependencies in fraud email classification.
- LSTM and Bi-LSTM significantly improved the False Negative rate, with only 42 and 55 misclassified fraud emails, respectively, making them highly effective in phishing detection.
- GRU also performed well, with only 28 false negatives and 57 false positives, suggesting that it is comparable to Bi-LSTM in terms of classification effectiveness.

The high performance of traditional models, particularly the SGD classifier, can be attributed to several key factors related to dataset characteristics and model suitability. First, the dataset used in this study, while moderately sized (17,538 emails), may not provide sufficient volume for deep learning models such as Bi-LSTM and GRU to fully leverage their capacity, often leading to overfitting. In contrast, traditional models like SGD and Logistic Regression are well-suited for smaller datasets and benefit from strong regularization techniques that enhance generalization. Second, the feature representation plays a critical role; traditional models utilized TF-IDF, a sparse and interpretable representation that captures term importance effectively in textual data, while deep models were trained on raw token sequences without leveraging pre-trained word embeddings. This absence of semantic prior knowledge limits the expressive power of deep models when trained from scratch on limited data. Additionally, the optimization characteristics of SGD make it particularly effective in high-dimensional text classification tasks, allowing for fast convergence and robust performance when combined with TF-IDF features. Together, these factors help to explain why traditional approaches, though simpler, outperformed more complex deep learning models in the context of fraud email detection.

To summarize the discussion on findings in confusion matrices, minimizing false negatives is crucial. A fraud email misclassified as safe (FN) is a significant risk, as it could allow cybercriminals to bypass security measures. Models like SGD, MLP, LSTM, and Bi-LSTM demonstrated strong FN reduction, making them highly suitable for real-world deployment. Balancing the false positives (FP) is also another critical point. While reducing FN is important, excessive FPs can overwhelm users with unnecessary security alerts. Traditional models like RF and MLP and deep learning models like GRU maintained a good balance, ensuring that legitimate emails are not excessively flagged. The Limitation of RNN is also another point to consider. The high FN and FP rates of RNN highlight its inefficiency in fraud email detection, reinforcing the need for LSTM, Bi-LSTM, or GRU, which are better suited for sequential text processing. The confusion matrices emphasize that SGD, MLP, LSTM, Bi-LSTM, and GRU are among the most effective models for phishing email detection due to their ability to minimize both False Negatives and False Positives. This highlights the importance of choosing models that ensure security without overwhelming users with excessive false alarms

The class-wise precision, recall, and F1-score analysis provides deeper insights into how well each classifier distinguishes between fraud emails and safe emails. These metrics are particularly important in fraud email detection, where the cost of misclassification can be high. The highest precision for the fraud class was achieved by SGD (0.9822), meaning that when the model predicted an email as fraud, it was correct 98.22 % of the time. This high precision indicates that SGD has a very low False Positive (FP) rate, reducing the risk of mistakenly flagging legitimate emails as fraud. It is an important point in avoiding unnecessary disruptions to users, ensuring that safe emails are not incorrectly blocked. The best precision for the safe class was observed in Random Forest (RF) with 0.9876, implying that 98.76 % of the emails classified as safe were indeed legitimate. This is critical because a high precision for safe emails ensures that fraud emails are not incorrectly labeled as safe (i.e., low FN). This makes RF particularly effective at preventing security

threats from fraud emails slipping through the detection system. The highest recall for the fraud class was achieved by MLP with 0.9843, indicating that 98.43 % of actual fraud emails were correctly identified. A high recall for fraud is also important in cybersecurity since a low recall would mean that many fraud emails remain undetected. MLP's high recall demonstrates its robustness in identifying phishing threats, reducing the chances of users falling victim to malicious emails. The highest recall for safe emails was observed in GRU (0.9866). It means that 98.66 % of safe emails were correctly classified. This finding ensures that legitimate emails are not mistakenly flagged as fraud, minimizing false alerts. GRU's high recall for safe emails makes it a strong candidate for real-world use, where user experience is highly considered because of the fact that excessive false alarms can reduce trust in the system. The best F1-score for fraud emails was obtained by SGD (0.9819). Since F1-score is the harmonic mean of precision and recall, this indicates that SGD achieves a strong balance between identifying fraud emails and minimizing misclassifications. The best F1-score for safe emails was recorded for GRU (0.9865). This shows that GRU performs well in balancing precision and recall for safe emails, proving that legitimate emails are correctly identified while minimizing misclassifications. The high F1-score also demonstrates that GRU is suitable for practical email filtering systems.

The selection of the optimal model is contingent upon the particular application or context. If prioritizing phishing detection, models with high recall for fraud, such as MLP, are preferable, whereas if minimizing false alarms is important, models with high precision for safe emails, such as RF, are better. To sum up, the study demonstrates that traditional machine learning methods like SGD and MLP, as well as deep learning approaches like GRU, offer highly effective solutions for fraud email detection. These models performed a balance between security and usability, providing that fraud emails are accurately detected while minimizing disruptions caused by false alarms. A detailed state-of-the-arts comparision (SOTA) table was provided by Table 9.

It is important to note that most state-of-the-art (SOTA) studies in fraud or phishing email detection utilize different datasets, often sourced from varied domains such as proprietary corporate collections or researcher-generated corpora. These datasets differ significantly in language style, structure, and labeling conventions, making direct performance comparisons unreliable and potentially misleading. Due to the absence of a universally accepted benchmark dataset in this domain, this study focuses on evaluating models on a consistent dataset to ensure internal validity. Nevertheless, representative SOTA results from recent literature are presented in Table 9 to provide contextual insight into current performance levels within the field.

8. Conclusions and future directions

In this study, a comprehensive analysis was conducted to evaluate the effectiveness of various machine learning and deep learning techniques in fraud email detection. The investigation involved traditional classifiers such as Naive Bayes (NB), Logistic Regression (LR), Stochastic Gradient Descent (SGD), XGBoost, Decision Tree (DT), Random Forest (RF), and Multi-Layer Perceptron (MLP), as well as deep learning models including Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Bi-Directional LSTM (Bi-LSTM), and Gated Recurrent Units (GRU). The models were trained on a publicly available dataset. TF-IDF feature extraction applied to transform email text into representations which is suitable for traditional classification while feature extraction step of text-to-sequence conversion was prefered for the deep learning approaches. The experimental results revealed that traditional machine learning classifiers outperformed deep learning models. SGD and MLP achieved the highest accuracy of 98.52 % and 98.43 %, respectively, while GRU, with an accuracy of 97.57 %, was the best-performing deep learning model. As future works suggestions, in addition to TF-IDF, incorporating semantic-based embeddings such as Word2Vec, BERT, or GPT-based models could enhance fraudulency detection by capturing deeper contextual meanings in emails. Combining traditional classifiers with deep learning models or employing stacked ensemble learning might further improve classification performance by leveraging the strengths of different models, but it will probably increase compuatational load. Future work could focus on deploying the best-performing models in a real-world email filtering system, analyzing their effectiveness over time, and incorporating feedback-based model improvements. Also, extending the study to non-English emails using multilingual NLP techniques would enhance global cybersecurity efforts since phishing attacks in its nature may occur in various languages.

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Table 9
The accuracy comparison of SOTAs to study in this paper.

Study	Method	Accuracy (%)
Mohammad RMA (Mohammad, 2024)	ELCADP	95.8
Gholampour et al. (Mehdi Gholampour and Verma, 2023).	K-Nearest Neighbor on a dataset generated by with GPT 2	94
Gangavarapu et al. (Gangavarapu et al., 2020).	Random forest with fi-based feature selection	98.4
Alhogail et al. (Alhogail and Alsabih, 2021).	Graph Convolution Network (GCN) and NLP techniques (tokenization, stop word removal)	98.2
Kumar et al. (Kumar and Sonowal, 2020).	Multinomial Naïve Bayes: using length, stemmer and hyperparameter tuning	98
Gupta et al. (Gupta et al., 2024).	BERT (feature extraction) + CNN (classifier)	97.5
The proposed model	TF-IDF + Stochastic Gradient Descent (SGD)	**98.52**

CRediT authorship contribution statement

Yunus Korkmaz: Writing – original draft, Visualization, Software, Methodology.

Declaration of competing interest

The author declares that there are no conflicts of interest related to this publication. Additionally, no substantial financial support which could have impacted the results or conclusions presented was received for this work that.

Data availability

Data will be made available on request.

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